

Midterm Report for Data Privacy: Membership Inference Attacks on Genomic Foundation Models

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Abstract

Machine learning models trained on genomic datasets introduce serious privacy concerns, particularly through membership inference attacks. In this type of attack, an adversary attempts to determine whether a specific individual’s genetic data was included in a model’s training dataset by analyzing prediction confidence and output patterns. Due to the fact that genomic data is uniquely identifying and highly sensitive, successful membership inference can reveal participation in medical studies or expose disease related genetic information. This report investigates the risk of membership inference in genomic foundation models, focusing on DNABERT-2 fine-tuned on tasks from the Genome Understanding Evaluation (GUE) benchmark. We analyze indicators of memorization such as the gap between training and testing accuracy and evaluate how model size and task constraints may increase vulnerability. We propose using shadow model techniques that replicate the target model’s behavior in order to detect confidence signatures associated with memorized data, specifically targeting the 70 bp Core Promoter Detection task. We also lay out the specific parameters we plan to vary, including LoRA rank, differential privacy noise multiplier (ϵ), and L2 regularization strength, and we describe how we will measure the privacy-utility tradeoff using attack accuracy, AUC-ROC, and downstream task MCC. Our expected results aim to classify genomic models based on their generalization gaps and identify conditions that increase memorization risk. The study also considers mitigation strategies such as differential privacy and regularization methods to reduce the likelihood of sensitive genomic data being inferred from trained models.

Keywords: Genomic Privacy, Membership Inference, Shadow Models, DNABERT-2, Differential Privacy, Regularization, GUE Benchmark

1 Introduction and Problem Statement

A membership inference attack on Machine Learning is a type of privacy attack where an adversary attempts to determine whether a particular person’s data was included in the training dataset of a machine learning model. In the context of genomic data, this attack is especially concerning because genetic information is extremely sensitive and uniquely tied to an individual. The attacker typically has access to the trained model and also possesses the genomic data of a

target individual. By querying the model with that person’s genomic features, the attacker analyzes the model’s output probabilities or confidence scores. Machine learning models often have higher confidence predictions for data they were trained on. By exploiting this difference, the attacker can infer whether the individual’s genome was part of the training set, potentially revealing that the person participated in a medical study or has a particular disease-related genetic profile.

This attack was first formalized by Shokri et al. (2017) [4], and it remains the standard framework that most follow-up work builds on. Their approach works in a black-box setting: the attacker does not need access to the model’s internal weights, just the ability to query it and observe the output prediction vector. The key insight is that ML models tend to behave differently on data they were trained on versus data they were not, because they have effectively memorized those examples to some degree. To exploit this, Shokri et al. introduced the *shadow model* technique, where the attacker trains their own set of proxy models that approximate the behavior of the target model. By labeling the shadow models’ training data as “member” and their test data as “non-member,” the attacker can train a binary classifier that learns to distinguish between the two based on the confidence signature of the target model’s outputs. They were able to demonstrate this attack against models hosted on Google and Amazon’s ML platforms, achieving high precision even without knowing the target model’s architecture. Since then, follow-up work like Ye et al. (2022) [5] has formalized this as a hypothesis testing problem and achieved even stronger attack performance using reference models.

The central research question we are trying to answer is: **To what extent are genomic foundation models, specifically DNABERT-2 fine-tuned on GUE benchmark tasks, vulnerable to membership inference attacks, and how effectively can differential privacy and regularization mitigate this risk while still keeping the model useful for downstream tasks?** This is an important question because genomic foundation models are increasingly being used in biomedical research, but their privacy properties have not really been examined. Our work is trying to address this gap.

1.1 Genomic Foundation Models and the GUE Benchmark

On the modeling side, recent years have seen the development of large-scale foundation models for genomic sequence analysis. DNABERT-2 (Zhou et al., 2024) [6] is one of the most prominent examples: a transformer-based model pre-trained on multi-species genome data with 117 million parameters. It introduced the Genome Understanding Evaluation (GUE) benchmark, which standardizes 36 classification tasks across 9 categories including core promoter detection, transcription factor prediction, and splice site prediction, with input lengths ranging from 70 to 10,000 base pairs. DNABERT-2 replaced the older k -mer tokenization with Byte Pair Encoding, which made it significantly more efficient while maintaining strong performance across the benchmark.

However, what has not been thoroughly examined is the privacy implications of training such a large model on genomic sequences. With 117 million parameters and some tasks using input sequences as short as 70 base pairs, there is substantial capacity for the model to store and recall specific training sequences rather than learning generalized features. This is the gap our work aims to address. We are specifically focusing on the fine-tuning stage, where task-specific genomic datasets are used to adapt the pre-trained model. This is the stage that is most relevant to real-world privacy risk, because fine-tuning is when sensitive, individual-level genomic data actually enters the model’s training pipeline.

1.2 Why Fine-Tuning Is the Critical Attack Surface

While pre-training is the foundation of models like DNABERT-2, attempting to use it for our membership inference attack would be too computationally expensive. The pre-training data is also generally public multi-species corpora, so there is less of a privacy concern there. Fine-tuning, on the other hand, typically involves curated datasets that may contain individual-level genomic sequences, for instance sequences associated with disease status or participation in a clinical study. Focusing on the fine-tuning stage is not only more feasible but actually provides a more accurate picture of how genomic data is exposed in practice. This is why our implementation focuses exclusively on the fine-tuning stage, which directly connects our methodology to the real-world privacy threat we are investigating.

2 Literature Review

2.1 Membership Inference Attacks

The foundational work by Shokri et al. (2017) [4] established the shadow model framework for membership inference. Their method trains multiple shadow models on data drawn from the same distribution as the target model’s training data. Each shadow model produces prediction vectors for its own training set (members) and a held-out set (non-members). These labeled prediction vectors are then used to train a binary attack classifier that learns to distinguish the confidence patterns that come from a model memorizing a data point versus generalizing. Ye et al. (2022) [5] extended

this by framing membership inference as a likelihood-ratio hypothesis test. Their approach computes the ratio of the target model’s confidence on a given sample to the confidence of a set of reference models, which gives a more principled statistical test and achieves stronger attack performance, especially against well-regularized models.

2.2 Differential Privacy as a Defense

Differential privacy helps defend against membership inference attacks by limiting how much influence any single individual’s data can have on a trained model [1]. In the context of machine learning, differential privacy is typically implemented via DP-SGD (Differentially Private Stochastic Gradient Descent), which clips per-sample gradients to a maximum norm and adds calibrated Gaussian noise before each parameter update. The privacy guarantee is parameterized by epsilon (ϵ): smaller values of ϵ provide stronger privacy but can weaken the predictive power of the model. It is important to find a balance in the amount of noise to ensure privacy guarantees and keep the model useful. Chen et al. (2020) [1] specifically studied DP in the context of genomic data and showed that DP-SGD can reduce the success rate of membership inference attacks on genomic classifiers, though at a non-trivial cost to classification accuracy. One key finding was that the optimal ϵ depends on the specific task and dataset size, so there is no universal value that works for all genomic applications. This is what motivates our plan to sweep across multiple ϵ values and measure the resulting tradeoff explicitly.

2.3 Memorization in Genomic Models

To motivate why we think memorization is a real concern, we point to Khan et al. (2025) [3] who introduced Hyperbolic Genome Embeddings and used homology-based splitting to test model generalization on the GUE benchmark. Their results show a significant performance reduction when models were tested on sequences that were biologically dissimilar to the training set. This tells us that models trained on the standard GUE splits may be memorizing sequence-level patterns rather than learning generalizable biological features. Ji et al. (2021) [2] also demonstrated with the original DNABERT that tasks with short input windows (such as the 70 bp Core Promoter Detection task) force models to rely on very subtle sequence signals, a condition that is highly conducive to memorization. This generalization gap signals to us that there’s a real issue of memorization going on here, and it directly supports our hypothesis that DNABERT-2, with its 117M parameters and access to short-sequence tasks, is at heightened risk of memorization and therefore vulnerable to membership inference.

3 Proposed Methods

In this section we lay out our complete implementation plan. We describe the threat model, the target task, how we will build the shadow models, how the attack classifier works, and the defense mechanisms we plan to evaluate. We also go through the specific parameters we will vary. Throughout

all of this, we are trying to maintain a clear connection back to our research question: how vulnerable is DNABERT-2 to membership inference, and how well can DP and regularization actually mitigate that?

3.1 Threat Model

We assume a black-box adversary who can query the fine-tuned DNABERT-2 model and observe the full output prediction vector (i.e., the softmax probabilities for each class). The adversary does not have access to the model’s internal weights or gradients. The adversary also has access to a dataset drawn from the same distribution as the target model’s fine-tuning data, which they use to train the shadow models. This is the standard threat model from Shokri et al. (2017) [4] and is realistic in the genomic context, because datasets like the GUE benchmark are publicly available and anyone can fine-tune DNABERT-2 on the same tasks.

3.2 Target Task: 70 bp Core Promoter Detection

We select the 70 bp Core Promoter Detection task from the GUE benchmark as our primary target. This task requires classifying whether a 70-base-pair DNA sequence contains a core promoter region. We chose this task for a few reasons. First, the short input length (70 bp) means the model has to rely on very subtle sequence-level signals, which increases the likelihood of memorization. Second, the task has a well-defined binary classification structure, which makes the attack classifier design simpler. Third, prior work [2, 3] has already documented generalization gaps on this task, which suggests memorization is already happening. We will also run our attack on a longer-context task as a secondary evaluation (e.g., Transcription Factor Prediction with 500 bp inputs) to see whether input length affects MIA vulnerability.

3.3 Fine-Tuning DNABERT-2 (Target Model)

We will fine-tune DNABERT-2 on the 70 bp Core Promoter Detection task using Low-Rank Adaptation (LoRA). LoRA inserts trainable low-rank matrices into the attention layers of the transformer, which allows for efficient fine-tuning without modifying the full 117M parameter set. We use LoRA rather than full fine-tuning both for computational feasibility and because LoRA is representative of how practitioners actually deploy these models in the real world. The specific fine-tuning hyperparameters we plan to use are shown in Table 1.

We vary the LoRA rank across $\{8, 16, 32\}$ to test whether increased fine-tuning capacity leads to more memorization and therefore higher MIA vulnerability. This is important because it lets us isolate the effect of model capacity on privacy risk, which is a core part of our research question.

3.4 Shadow Model Construction

Following Shokri et al. (2017) [4], we train a set of shadow models that mimic the target model’s behavior. The way this works is as follows:

Step 1 – Data partitioning. We take the full GUE Core Promoter Detection dataset and split it into disjoint subsets.

Table 1: Fine-Tuning Hyperparameters for Target Model

Parameter	Value(s)
Base model	DNABERT-2 (117M params)
Adaptation method	LoRA
LoRA rank (r)	8, 16, 32
LoRA alpha	$2 \times r$
Target modules	Query & Value projections
Learning rate	2×10^{-5}
Batch size	32
Epochs	10
Optimizer	AdamW
Train/test split	80/20 stratified

The target model is trained on subset D_{target} . Each shadow model i is trained on a different subset $D_{\text{shadow},i}$ drawn from the remaining data. We make sure that D_{target} and each $D_{\text{shadow},i}$ are disjoint but drawn from the same distribution.

Step 2 – Shadow model training. Each shadow model is fine-tuned using the same architecture (DNABERT-2 + LoRA) and the same hyperparameters as the target model. This is important because it ensures the shadow models’ behavior approximates the target model’s behavior. We plan to train $N = 4$ shadow models, which we think balances computational cost with having enough data for the attack classifier.

Step 3 – Labeling. For each shadow model, we query it on (a) its own training data (labeled “member”) and (b) a held-out set (labeled “non-member”). We record the full softmax output vector for each query. This gives us a labeled dataset of (prediction vector, member/non-member) pairs.

Step 4 – Attack classifier training. We aggregate the labeled data from all shadow models and train a binary classifier (either logistic regression or a small MLP) to predict membership from the prediction vector. We also plan to compute the normalized entropy of each prediction vector as an additional feature, since low entropy indicates high confidence and may signal memorization.

3.5 Executing the Attack

Once we have the attack classifier trained, we evaluate it on the actual target model. We query the target model on (a) its actual training set D_{target} (true members) and (b) a held-out set that was not used for training (true non-members). The attack classifier predicts membership for each sample. We evaluate how well the attack works using the metrics described in Section 4.

3.6 Defense Mechanisms

To evaluate mitigation strategies, we re-train the target model under two different defense regimes and re-run the full attack pipeline for each one. This lets us directly measure how each defense affects both the MIA vulnerability and the downstream task performance.

3.6.1 Differential Privacy (DP-SGD).

We plan to apply DP-SGD during fine-tuning using the Opacus library. DP-SGD clips per-sample gradients to a maximum norm C and adds Gaussian noise scaled by the noise multiplier σ before each parameter update. The resulting privacy guarantee is expressed as (ϵ, δ) -DP, where ϵ is the privacy budget and δ is a negligible failure probability (we set $\delta = 1/n$ where n is the training set size). We will sweep the noise multiplier σ across $\{0.5, 1.0, 1.5, 2.0\}$ and the gradient clipping norm C across $\{0.5, 1.0\}$, and then compute the corresponding ϵ values using Opacus’s privacy accountant. The specific parameters are shown in Table 2.

Table 2: Differential Privacy Parameters

Parameter	Value(s)	Purpose
Noise mult. (σ)	0.5, 1.0, 1.5, 2.0	DP noise magnitude
Clip norm (C)	0.5, 1.0	Gradient sensitivity
δ	$1/n$	Failure probability
ϵ (computed)	Target: 1–10	Privacy budget
Epochs	10	Cumulative cost

3.6.2 L2 Regularization (Weight Decay).

L2 regularization adds a penalty proportional to the squared magnitude of the model’s weights to the loss function. This discourages the model from assigning too much value to any specific weight, which can reduce memorization by smoothing the model’s decision boundaries. We will sweep the weight decay coefficient λ across $\{0.001, 0.01, 0.05, 0.1\}$. Because the model parameters are so large (117M), we hypothesize that weight decay will meaningfully decrease memorization, though it will probably come at some cost to task performance. We apply weight decay through the AdamW optimizer’s built-in weight decay parameter.

3.6.3 Combined Defense.

We also want to evaluate a combined defense that applies both DP-SGD and L2 regularization at the same time. We will take the best-performing individual ϵ and λ values (based on the privacy-utility tradeoff) and combine them. This tests whether the two defenses are complementary or if they just compound the utility cost without giving proportional privacy gains.

4 Evaluation Metrics and Privacy-Utility Tradeoff

One of the most important parts of our study is measuring the privacy-utility tradeoff: how much task performance do you have to give up for each unit of privacy gained? We use the following metrics to measure this.

4.1 Attack Success Metrics (Privacy)

Attack Accuracy measures the fraction of samples for which the attack classifier correctly predicts membership status. A random-guess baseline gives you 50% accuracy, so values significantly above 50% indicate a successful attack. **AUC-**

ROC (Area Under the Receiver Operating Characteristic Curve) gives us a threshold-independent measure of how well the attack classifier can discriminate between members and non-members. An AUC of 0.5 means no better than random guessing, while 1.0 means perfect discrimination. We also plan to report **TPR@FPR=0.1%** (True Positive Rate at a low False Positive Rate), which measures how well the attack works in a realistic setting where the attacker wants high confidence. This is the metric that Carlini et al. recommend for evaluating MIA strength.

4.2 Task Performance Metrics (Utility)

Matthew’s Correlation Coefficient (MCC) is the standard metric used by the GUE benchmark and is appropriate for binary classification tasks that may have class imbalance. MCC ranges from -1 to $+1$, where $+1$ indicates perfect classification. We also report **test accuracy** for comparability with prior work. The **train-test accuracy gap** (training accuracy minus test accuracy) serves as a proxy for how much memorization is going on: a large gap means the model has memorized training data rather than learning features that generalize.

4.3 Measuring the Tradeoff

For each experimental configuration (baseline, DP with various σ , L2 with various λ , and combined), we record both the attack metrics and the task performance metrics. We then plan to plot (a) attack AUC vs. task MCC and (b) attack accuracy vs. ϵ value. A good defense would be one that substantially reduces attack AUC while keeping MCC at an acceptable level. We define “acceptable” as MCC within 10% of the undefended baseline, following the convention from Chen et al. (2020) [1]. Of course, after we evaluate memorization after regularization, we also want to evaluate actual data quality to make sure that too much useful information isn’t lost. This framework lets us directly answer our research question by identifying what parameter settings give you the best privacy-utility balance for genomic foundation models.

5 Experimental Design Summary

Table 3 summarizes all the experiments we plan to run. Each row is a different experimental configuration. For every configuration, we run the full pipeline: fine-tune the target model, train shadow models with the same configuration, train the attack classifier, and evaluate both attack success and task performance.

6 Expected Results and Hypotheses

Based on our literature review and what we know about DNABERT-2, we came up with the following hypotheses:

H1 (Baseline vulnerability): The undefended DNABERT-2 fine-tuned on the 70 bp Core Promoter Detection task will have MIA attack accuracy significantly above 50% (we expect somewhere in the 65–80% range) and AUC above 0.7, which would indicate meaningful privacy leakage.

Table 3: Experimental Configurations

Config	LoRA Rank	DP (σ , C)	Weight Decay (λ)	Purpose
Baseline	8, 16, 32	None	0 (default)	Measure undefended vulnerability
DP-only	16	$\sigma \in \{0.5, 1.0, 1.5, 2.0\}$, $C \in \{0.5, 1.0\}$	0	Isolate DP effect
L2-only	16	None	$\lambda \in \{0.001, 0.01, 0.05, 0.1\}$	Isolate regularization effect
Combined	16	Best σ , C from DP-only	Best λ from L2-only	Test complementarity
Long-context	16	None	0	Compare 500 bp vs. 70 bp

H2 (Capacity effect): Higher LoRA rank ($r = 32$) will yield higher attack accuracy than lower rank ($r = 8$), because greater fine-tuning capacity gives the model more room to memorize.

H3 (DP effectiveness): DP-SGD with noise multiplier $\sigma \geq 1.0$ will reduce attack AUC to below 0.6, but at a cost of 5–15% reduction in task MCC depending on the ε achieved.

H4 (L2 effectiveness): L2 regularization with $\lambda \geq 0.01$ will reduce the train-test accuracy gap and provide a modest reduction in attack accuracy (around 5–10 percentage points), but will be less effective than DP at actually reducing MIA vulnerability.

H5 (Context length): The 70 bp task will be more vulnerable to MIA than the 500 bp task, because shorter sequences force the model to rely more on memorization of specific sequence patterns.

H6 (Combined defense): Combining DP and L2 will give the strongest privacy protection, but the marginal improvement over DP alone may not be worth the additional hit to utility.

7 Implementation Timeline and Next Steps

We have organized our remaining work into three phases. In **Phase 1 (Weeks 1–2)**, we will fine-tune the baseline target model on the 70 bp Core Promoter Detection task using LoRA ranks 8, 16, and 32, and record train/test accuracy and MCC for each configuration. In **Phase 2 (Weeks 2–3)**, we will construct the shadow models, train the attack classifier, and evaluate the baseline MIA vulnerability. In **Phase 3 (Weeks 3–4)**, we will re-run the pipeline with the DP-SGD defenses (sweeping σ and C), L2 regularization (sweeping λ), and the combined defense. We will also run the secondary experiment on the 500 bp task. All results will be compiled into privacy-utility tradeoff plots and a final report.

We plan to use the CWRU HPC cluster for computational resources. We estimate roughly 20–30 GPU-hours for the full experimental pipeline, including all configurations and shadow model training.

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